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Alterations in lysosomal and proteasomal markers in Parkinson's disease: Relationship to alpha-synuclein inclusions

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ABSTRACT

We explored the relationship between ubiquitin proteasome system (UPS) and lysosomal markers and the formation of α-synuclein (α-syn) inclusions in nigral neurons in Parkinson disease (PD). Lysosome Associated Membrane Protein 1(LAMP1), Cathepsin D (CatD), and Heat Shock Protein73 (HSP73) immunoreactivity were significantly decreased within PD nigral neurons when compared to age-matched controls. This decrease was significantly greater in nigral neurons that contained α-syn inclusions. Immunoreactivity for 20S proteasome was similarly reduced in PD nigral neurons, but only in cells that contained inclusions. In aged control brains, there is staining for α-syn protein, but it is non-aggregated and there is no difference in LAMP1, CatD, HSP73 or 20S proteasome immunoreactivity between α-syn positive or negative neuromelanin-laden nigral neurons. Targeting over-expression of mutant human α-syn in the rat substantia nigra using viral vectors revealed that lysosomal and proteasomal markers were significantly decreased in the neurons that displayed α-syn-ir inclusions. These findings suggest that α-syn aggregation is a key feature associated with decline of proteasome and lysosome and support the hypothesis that cell degeneration in PD involves proteosomal and lysosomal dysfunction, impaired protein clearance, and protein accumulation and aggregation leading to cell death.

Introduction

In recent years, converging evidence suggests that the accumulation of unwanted and misfolded proteins, due to increased protein production and/or impaired clearance, plays a central role in the pathogenesis in PD (Olanow, 2007). PD is characterized by the accumulation of aggregated proteins in Lewy bodies and Lewy neurites. Genetic studies (see review: Bonifati, 2007) have identified familial cases that are associated with mutations that could cause excess production of mutant or wildtype proteins (e.g. α-synuclein) (Polymeropoulos et al., 1997; Singleton et al., 2003) or impairment in the capacity of the UPS or lysosomal systems to clear unwanted proteins (e.g. Parkin, LRRK2, DJ-1, and PINK1 mutation) (Abou-Sleiman et al., 2006; Wang et al., 1999; Chung et al., 2004). Further, pathological studies demonstrate that there are alterations in proteasomal structure and function in the substantia nigra pars compacta (SNc) in patients with sporadic PD (McNaught et al., 2003; Bedford et al., 2008).

Much of the interest in the role of misfolded proteins in PD has focused on alpha-synuclein (α-syn). α-syn is a major component of Lewy bodies in sporadic forms of PD (Spillantini et al., 1997) and mutations or multiplication of the gene are associated with rare familial forms of the disorder (Kruger et al., 1998; Polymeropoulos et al., 1997; Singleton et al., 2003; Nishioka et al., 2006). Over-expression of mutant or wildtype α-syn in transgenic (tg) mice (Masliah et al., 2000) and *Drosophila* (Feany and Bender, 2000) can cause alterations in nigrostriatal function and in some instances dopamine neuronal degeneration (e.g. Thiruchelvam et al., 2004) with inclusion bodies. Similarly targeted over-expression of α-syn in the nigrostriatal system of rats and nonhuman primates (Kirik et al., 2002; 2003) induces neuronal degeneration and mimics several of the pathological and behavioral features of PD. The link between α-syn and neurodegeneration is further strengthened by the finding that accumulation of α-syn is associated with down-regulation of dopamine phenotypes in the SNc with aging of normal humans and monkeys (Chu and Kordower, 2007) and in PD (Chu et al., 2006). Down-regulation of alpha-synuclein expression can rescue dopaminergic cells from cell death in the SNc of Parkinson's disease model (Hayashita-Kinoh et al., 2006).

Although the precise mechanism of degradation is not precisely understood, clearance of α-syn appears to involve both the UPS and lysosome systems (Pan et al., 2008; Vogiatzi et al., 2008; Bennett et al., 2005; Olanow and Mcnaught, 2006). Lysosomes are cellular organelles that contain a wide variety of proteases (e.g. cathepsins and hydrolases) whose activity maintains the turnover of membrane

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proteins and other cellular macromolecules. Lysosomal processing capacity diminishes progressively over the lifespan of an animal (Kiffin et al., 2007; Terman, 2006), and lysosomal malfunction is associated with age-related neurodegenerative disorders (Nixon et al., 2000; Rubinsztein, 2006; Terman, 2006; Bandhyopadhyay and Cuervo, 2007; Pan et al., 2008). The connection between lysosomal dysfunction and neurodegeneration is further demonstrated by the finding that inhibition of lysosomal enzymes leads to abnormal protein accumulation and aggregation, synaptic loss, and neuronal demise in laboratory models (Ivy et al., 1984; Koike et al., 2000; Bendiske and Bahr, 2003; Felbor et al., 2002). The UPS is the primary system for clearing unwanted intracellular proteins from eukaryocytes (Goldberg, 1999; Tanaka et al., 2001a). Misfolded proteins are poly-ubiquitinated at lysine residues which signal for their transfer to, and degradation by, the proteasome. The proteasome is a multicatalytic protease with trypsin, chymotrypsin, and PHPG activities that degrade the unwanted proteins into their constituent amino acids and small peptides. In the PD nigra, abnormalities in the expression of the alpha subunit of the proteasome, and in each of its enzyme activities, have been observed (McNaught et al., 2003). Further, proteasomal inhibition in laboratory models leads to degeneration of dopamine neurons coupled with inclusions that stain for α-syn (McNaught et al., 2006; Rideout et al., 2005; Lim, 2007). The purpose of the present study was to assess the interrelationships between α-syn accumulation, aggregation and markers of the lysosome and UPS systems in PD. To do this, we evaluated 1) whether lysosomal and UPS markers are altered in remaining substantia nigra neurons in PD; 2) whether the formation of α-syn inclusions in PD is specifically associated with disturbances in markers of the lysosome and UPS; and 3) whether viral over-expression of α-syn is, by itself, sufficient to induce changes in markers of lysosomal and UPS function in the rodent substantia nigra. To accomplish these aims, the relative levels of proteins involved in UPS and lysosomal functional pathways were analyzed by immunostaining and with the use of quantitative immunofluorescence intensity measurements within the substantia nigra of PD cases and in rats with targeted over-expression of α-syn. To determine the specificity of any findings, the data obtained in these cases were compared with findings seen in age-matched controls.

Materials and methods

Human tissue acquisition and processing

Tissues from 20 subjects were examined in this study. They either had a clinical and neuropathological diagnosis of PD ($n = 10$) or were normal age-matched controls ($n = 10$). All cases were used for all analyses. The number of cells per case that were analyzed were as follows: 100 nigral cells in normal cases; 50–70 nigral cells/PD case that contained inclusion and 80–100 nigral cells/PD case that did not contain inclusions. PD cases were all evaluated at the Movement Disorder Center at Rush University and were diagnosed according to standard clinical and neuropathological criteria (see Chu et al., 2006). Control subjects were participants in the Religious Order Study, a longitudinal clinical pathological study of aging (Chu et al., 2002, 2006) and were without neurological or psychiatric illnesses during life or neuropathological abnormalities at postmortem. The study was approved by the Human Investigation Committee at Rush University Medical Center. There were no significant differences in age at time of death or postmortem interval among the two groups examined (Table 1).

At autopsy, brains were removed from the calvaria and processed as described previously (Chu et al., 2002; Chu and Kordower 2007). Briefly, each brain was cut into 1 cm thick coronal slabs using a Plexiglas brain slice apparatus and then hemisected. The brains were then immersion-fixed in 4% paraformaldehyde for 48 h at 4°C and

Table 1
Summary of case demographics.

	Age (yr)	Gender	PMI (h)	Braak score	MMSE
Age-matched control cases					
1	74	M	10.3	n/a	27
2	73	M	7.3	I/II	28
3	4	M	4.0	I/II	28
4	83	M	5.5	V/IV	26
5	72	F	7.0	III/IV	30
6	91	F	10.7	III/IV	27
7	86	F	5.0	n/a	30
8	88	F	7.0	III/IV	28
9	70	M	6.0	II/III	30
10	68	M	6.5	III/IV	28
Mean ± SE	78.90 ± 8.34		6.93 ± 2.14		28.20 ± 1.39
	Age (yr)	Gender	PMI (h)	UPDRS III	H and Y stage
Parkinson's disease cases					
1	73	M	5.5	27	3
2	73	M	13.0	n/a	n/a
3	77	M	4.0	49	3
4	72	M	5.8	23	2
5	78	M	4.0	64	5
6	76	F	3.3	54	4
7	81	M	3.1	54	5
8	63	F	5.0	47	4
9	72	M	4.0	42	3
10	76	F	8.0	65	5
Mean ± SE	74.10 ± 4.86		5.57 ± 2.98	47.22 ± 14.64	3.77 ± 1.09
	Age (yr)		Gender		PMI (hr)

PMI, postmortem interval; n/a, not available; MMSE, mini-mental status examination; UPDRS, United Parkinson's Disease Rating Scale; H and Y, Hoehn and Yahr.

cryoprotected in 0.1 M phosphate buffer saline (PBS; pH 7.4) containing 2% dimethyl sulfoxide (DMSO), 10% glycerol for 24 h followed by 2% DMSO, 20% glycerol in PBS for at least 2 days before sectioning. Slabs containing substantia nigra were cut perpendicular to the longitudinal axis of the brainstem into 18 adjacent series of 40-μm thick sections on a freezing sliding microtome. All sections were collected and stored sequentially in a cryoprotectant solution before processing.

Animals and surgery

Young adult Sprague Dawley rats (Charles River Laboratories, Wilmington, MA) were housed two to a cage with ad libitum access to food and water during a 12 h light/dark cycle, according to the rules set by the Research Ethical Committee at Rush University Medical Center. Recombinant adeno associated virus serotype 6 vector encoding human mutant (A30P) α-syn gene (rAAV6-h-A30P) were prepared and titered as described previously (Towne et al., 2008). Under halothane anesthesia, 2 μl of the vector suspension was injected stereotaxically into the right nigral region at 5.3 mm posterior and 2.3 mm lateral to bregma, 7.7 mm ventral to dura. The needle was kept in place for an additional 5 min before slowly being withdrawn. At 6 weeks after injection, animals (rAAV6-h-A30P, $n = 10$) were perfused through the ascending aorta with physiological saline, followed by 4% ice-cold paraformaldehyde. The brains were post-fixed in the same solution for 2 h, transferred to10%, 20%, 30% sucrose, and sectioned on a freezing microtome at 40 μm in the coronal plane. All sections were collected and stored in order in a cryoprotectant solution before processing. All animal experiments had prior approval from the Institutional Animal Care and Use Committee at Rush University.

Antibodies

In order to assess the integrity of the lysosomal and UPS pathways, antibodies were selected to examine 1) the lysosomal associated membrane protein 1 (LAMP1), 2) the lysosomal protease cathepsin D (CatD), 3) the constitutive chaperone protein heat shock protein (HSP73), a combinatorial member of the HSP70 family of proteins, and 4) the human 20s proteasome. LAMP1 (ab 24170, abcam, MA) is a rabbit polyclonal antibody recognizing a band of approximately 120 kDa in Western blots (Sarafian et al., 2006). The CatD (ab 826, abcam, MA) is a rabbit polyclonal antibody against the full length of CatD and reacts with zymogen as well as the activated form of CatD (Henry et al., 1990). HSP73, a rat monoclonal antibody (MAb1B5, Stressgen, MI) detects a ~73 kDa protein under native and denaturing condition (Andringa et al., 2006) and a rabbit polyclonal antibody (ab301010, abcam, MA) corresponding to amino acids 574–591 of human HSP70. The 20S proteasome was a rabbit polyclonal antibody (ab3325, abcam, MA) directed at amino acid residues 249–263 of the C-terminus of human proteasome 20S C2 subunit. The 20S proteasome antibody recognizes a single band at 29 kDa in Western blots (Dutaud et 2002). To assess the α-syn expression, the LB509 antibody was employed. LB509 (mouse monoclonal antibody; Zymed Lab, San Francisco, CA) directly recognizes amino acids 115–122 of human α-syn (P37840; Jakes et al., 1999; Giasson et al., 2000). Results were confirmed with a second antibody to α-syn (ab15530, rabbit polyclonal antibody; abcam, MA) which recognizes C-terminal amino acids 111–131 of human α-syn (Choi et al., 2006).

To control for antibody specificity, adsorption experiments were performed for LAMP1, HSP73, and 20S proteasome antibodies. Briefly, these antibodies were combined with a five fold volume (by weight) of blocking peptides (ab25744 for LAMP1, ab4943 for 20S proteasome, Abcam; HPP751 for HSP, StressGen) separately in TBS and incubated overnight at 4°C. The immune complexes with the antibody and blocking peptide were centrifuged at 10,000 rpm for 20 min. The adsorbed protein/antibody complex was then used in lieu of the primary antibody. Staining specificity was further confirmed by omitting the primary antibody (which control for the specificity of the staining procedure and the secondary antibody) and replacement of the primary antibody with an irrelevant IgG matched for protein concentration. All of these control experiments resulted in a total absence of staining.

Double label immunofluorescence

A double label immunofluorescence procedure was employed in age-matched controls, PD cases and rats with targeted over-expression of α-syn to assess whether the lysosomal and proteasomal markers were altered in nigral neurons that specifically co-expressed α-syn. Sections through the substantia nigra from each brain were incubated in the first primary antibody overnight at room temperature and goat-anti-rabbit or goat-anti-rat antibodies coupled to Cy5 (1:200; Jackson ImmunoResearch, West Grove, PA) for 1 h. Following six washes in TBS, the sections were blocked again for 1 h in a solution containing 5% serum, 2% bovine serum albumin, and 0.3% Triton X100 in TBS. Sections were then incubated in α-syn antibody overnight and goat-anti-mouse antibodies coupled to Cy2 (1:200; Jackson ImmunoResearch, West Grove, PA) for 1 h. The sections were mounted on gelatin-coated slides and air dry overnight. To block auto-fluorescence the sections were rinsed in distil water, dehydrated in 70% alcohol for 5 min, incubated in the auto-fluorescent eliminator reagent (2160; Chmicon) for 5 min, and immersed in three changes of 70% alcohol. After rinsing in disttil water, the sections were coverslipped using polyvinyl alcohol with DABCO (10981, Sigma, St. Louis, mo). The images were scanned using Olympus Confocal Fluoroview Microscope. Auto-fluorescence represents a potentially confounding factor using immunofluorescence techniques in this tissue. However, we employed two means to minimize auto-fluorescence effectively. First, the fluorescence CY5 is a fluorophore in the infrared range and the auto-fluorescence from lipofuscin is minimized within this range of the visual spectrum. Second, use of the auto-fluorescent eliminator reagent further blocked auto-fluorescence in the tissue. The images were scanned using Olympus Confocal Fluoroview Microscope.

Fluorescence optical densitometry

To measure the relative immunoreactive (-ir) intensity in nigral neurons, all immunofluorescence double labeled images were scanned with the Olympus Confocal Fluoroview microscope equipped with argon and krypton lasers. The laser intensity, confocal aperture, PMT voltage, offset, electronic gain, scan speed, image size, filter, and zoom were set for the background level with a control section and maintained throughout the entire experiment to maintain consistency of the scanned image for every slide (Chu and Kordower, 2007; Chu et al., 2002). Imaging was performed with a 20× objective and a 488 or 647 nm excitation source. All optical density (OD) measurements were performed using stereological principles of random and systematic sampling. More than 100 cells were valuated for each population in each marker. The intensity mapping sliders ranged from 0 to 4,095; 0 represented a maximum black image, and 4,095 represented a maximum bright image. The LAMP1-ir, CatD-ir, HSP73-ir or 20S proteasome-ir neurons with or without α-syn-ir profiles were identified and outlined separately by an investigator blinded to the clinical and pathological data. Quantitative OD of immunofluorescence intensity measurements were performed on individual neuromelanin (NM) laden neurons irrespective of their α-syn profiles using FLUOVIEW software. For each marker, five equispaced sections across the entire length of the substantia nigra were sampled and evaluated. To account for differences in background staining intensity, five background intensity measurements lacking immunofluorescent profiles were taken from each section. The mean of these five measurements constituted the background intensity. The background intensity was then subtracted from the measured immunofluorescence intensity of each individual neuron to provide a final immunofluorescence intensity value.

To confirm co-localization of lysosomal and proteasomal markers with α-syn immunofluorescence, optical scanning through the neuron's *z*-axis was performed at 1 μm thicknesses, and neurons suspected of being double labeled were analyzed in the cross-section view through the *z*-axis to ensure the accuracy of this observation.

Proteinase K treatment and immunostaining

To determine whether the α-syn seen in nigral neurons was soluble (non-aggregated) or insoluble (aggregated), midbrain sections from rAAV6-h-A30P rats, age-matched controls and PD patients were digested with proteinase K (PK; Chu and Kordower, 2007). Briefly, sections containing the substantia nigra were mounted onto gelatin-coated slides and dried for at least 8 h at 55°C. After wetting with TBST (10 mM Tris–HCl, pH 7.8; 100 mM NaCl; 0.05% Tween20), the sections were digested with 50 μg/ml PK (Invitrogen) in TBST (10 mM Tris–HCl, pH 7.8; 100 mM NaCl; 0.1% Tween20) for a period of 1.5 h at 55°C. The sections were fixed with 4% paraformaldehyde for 10 min. After several washes, the sections were then processed for α-syn immunostaining.

Data analysis

The OD measurements were analyzed using a factorial analysis of variance model (ANOVA). All statistical tests used α level set at 0.05 (two-tailed). When appropriate, post hoc comparisons between groups were performed by using the method of Scheffe' to control for multiple comparisons.

Digital illustrations

Confocal images were exported from the Olympus Laser Scanning Microscopy Fluoview and stored as TIFF files. Conventional light microscopic images were acquired using a Nikon Microphoto FXA Microscope attached to a Nikon digital camera DXM1200 and stored as TIFF files. All figures were prepared using Photoshop 8.0 graphics software. Only minor adjustments in brightness were made.

Results

Lysosomal and proteasomal marker immunoreactivity in normal controls: Qualitative observations

In the aged normal control group, numerous NM-laden nigral neurons displayed diffuse non-aggregated α-syn-ir staining throughout the cytoplasm (Figs. 1C, 2C, 3C, 4C, 5A) that was digestible by proteinase K (Fig. 5B), as we have previously reported (Chu and Kordower, 2007). Immunohistochemistry revealed that LAMP1-ir (Fig. 1A), CatD-ir (Fig. 2A), HSP73-ir (Fig. 3A) and 20S proteasome-ir (Fig. 4A) neurons were widely distributed throughout the SNc in all normal control brains. The LAMP1 and CatD immunostaining profiles were characterized by cytoplasmic granules diffusely distributed in NM-laden neurons (Figs. 1B, 2B, 3B, 4B). Intense LAMP1 and CatD immunoreactivity was also observed in scattered NM negative small cells (Figs. 1A, 2A). HSP73-ir was less granular and observed in both the perikarya and processes of NM-laden neurons (Fig. 3A). 20S proteasome immunostaining was mainly detected within the nucleus of nigral neurons (Fig. 4A).

In order to determine whether alterations in lysosomal and proteasomal markers were associated with aberrant accumulation of the α-syn protein, we performed confocal microscopy using dual staining for α-syn and each of the lysosomal and proteasomal markers described above. Co-localization analysis in the normal human aging nigra revealed that there was no apparent differences in the immunofluorescent intensities of LAMP1 (Fig. 1D), CatD (Fig. 2D), HSP73 (Fig. 3D) or 20S proteasome (Fig. 4D) in α-syn-ir positive and negative NM-laden nigral neurons. It should be noted that α-syn accumulation in controls was diffuse (Figs. 1C–4C, 5A) and not associated with inclusion body formation. These results indicate that the accumulation of non-aggregated α-syn is not associated with alterations in staining for lysosomal and proteasomal markers in the normal human aged brain.

Lysosomal and proteasomal markers in PD: Qualitative observations

Immunoreactivity for lysosomal and proteasomal markers in PD were easily distinguishable from age-matched controls. Robust decreases of LAMP1 (Fig. 1E), CatD (Fig. 2E), and HSP73 (Fig. 3E) staining within individual NM-laden nigral neurons (Figs. 1F–4F) were observed in all PD brains. In contrast, immunostaining for the 20S proteasome was variable, with some cells displaying marked decreases in immunoreactivity while others appeared normal (Fig. 4E). Indeed, many NM-laden nigral neurons in PD showed intense 20S proteasome immunoreactivity.

In order to evaluate whether the α-syn inclusions were associated with altered expression of lysosomal and proteasomal markers, co-localization studies were performed. α-syn-ir inclusions labeled with the LB509 antibody were detected in a subset of NM-laden nigral neurons throughout the SNc in all PD cases (Figs. 1F–4F, 5C). Co-localization analyses revealed that LAMP1 (Fig. 1H) CatD (Fig. 2H), HSP73 (Fig. 3H), and 20S proteasome (Fig. 4H) immunostaining were significantly lower in neurons that contained α-syn-ir inclusions.

Lysosomal and proteasomal markers in substantia nigra in PD and age-matched controls: Quantitative observations

We used optical densitometry to quantitatively assess relative levels of each marker in PD and aged matched controls. Since α-syn inclusions occupy intracellular spaces that could affect OD measures, we quantified ODs only in cellular regions that did not contain this structure. We performed fluorescence intensity measurements on individual neurons immunostained for lysosomal and proteasomal markers (Table 2). In PD patients, there were significant reductions in OD measure for LAMP1 ($F[1, 18]=37.31$; $P<0.001$), CatD ($F[1, 18]=11.54$; $P<0.001$), HSP73 ($F[1, 18]=9.42$; $P<0.001$), and 20S proteasome ($F[1, 18]=7.93$; $P<0.05$).

In order to understand whether decreases in ODs of the lysosomal and proteasomal markers were associated α-syn positive inclusions in PD, we analyzed the relative intensities of LAMP1, CatD, HSP73, and 20S proteasome in nigral neurons that did or did not contain α-syn inclusions. We demonstrated above that in normal controls, the OD of all lysosomal and proteasomal markers were similar in nigral neurons independent of whether they were α-syn immunonegative or whether they displayed soluble (proteinase K digestable) α-syn. In PD, however, NM-laden neurons that do not display any α-syn are rare. Thus in the present analysis, the relative levels of lysosomal and proteasomal markers in nigral neurons were made between 1) nigral cells from age-matched controls irrespective of α-syn profile, 2) PD nigral neurons that displayed soluble (proteinase K digestable) α-syn and 3) nigral cells that displayed aggregated (proteinase K nondigestable).

In PD, OD measurements of LAMP1, CatD, and HSP73 immunoreactivity were significantly diminished in comparison with age-matched controls in neurons that contain either non-aggregated α-syn-ir or that contained α-syn-ir aggregates ($P<0.01\sim0.001$; Figs. 6A–C). However, the reduction in OD for LAMP1, CatD, and HSP73 immunoreactivity was significantly more pronounced in nigral neurons displaying α-syn-ir positive aggregates compared with PD cells that contained non-aggregated α-syn-ir $P<0.05\sim0.01$; Figs. 6A–C). OD measurements for 20S proteasome were also significantly decreased, but only in nigral neurons displaying α-syn-ir positive aggregates ($P<0.001$; Fig. 6D). Nigral neurons that lacked α-syn-ir displayed 20S proteasome immunofluorescence OD measurements that were similar to age-matched controls (Fig. 6D).

Targeted viral over-expression of α-syn in the rat substantia nigra: Qualitative observations

α-syn-ir was undetectable in rats on the non-injected side. In contrast, an intense expression of human α-syn protein was observed in the substantia nigra and striatum of rats that received rAAV6-h-A30P (Fig. 5E). Several distinct staining patterns of α-syn were observed in nigral neurons ipsilateral to the rAAV6-h-A30P injection. In some nigral neurons, α-syn-ir was diffusely distributed throughout the cell including the soma and proximal dendrites, with no visible inclusions. In others, dense α-syn-ir cytoplasmic aggregates (Figs. 5E, 7B, 8B, 9B, 10B) were seen that resembled Lewy body's seen in PD (Figs. 5E, F). α-syn-ir swollen, dystrophic neurites were also observed in the substantia nigra and striatum. PK digestion was used to ascertain whether the over-expression of α-syn led to accumulations that were soluble (non-aggregated) or insoluble (aggregated). Following PK treatment, the dense aggregates of α-syn-ir remained (Fig. 5F), whereas the α-syn-ir was eliminated from neurons that displayed diffuse α-syn cytoplasmic accumulation (Fig. 5E). This compares with PD where α-syn stained inclusions and neurites can persist following PK digestion (Fig. 5D), while diffuse α-syn-ir is eliminated. These results indicate that rAAV6-h-A30P in rat leads to insoluble α-syn aggregation that mimics the neuropathological findings in PD.

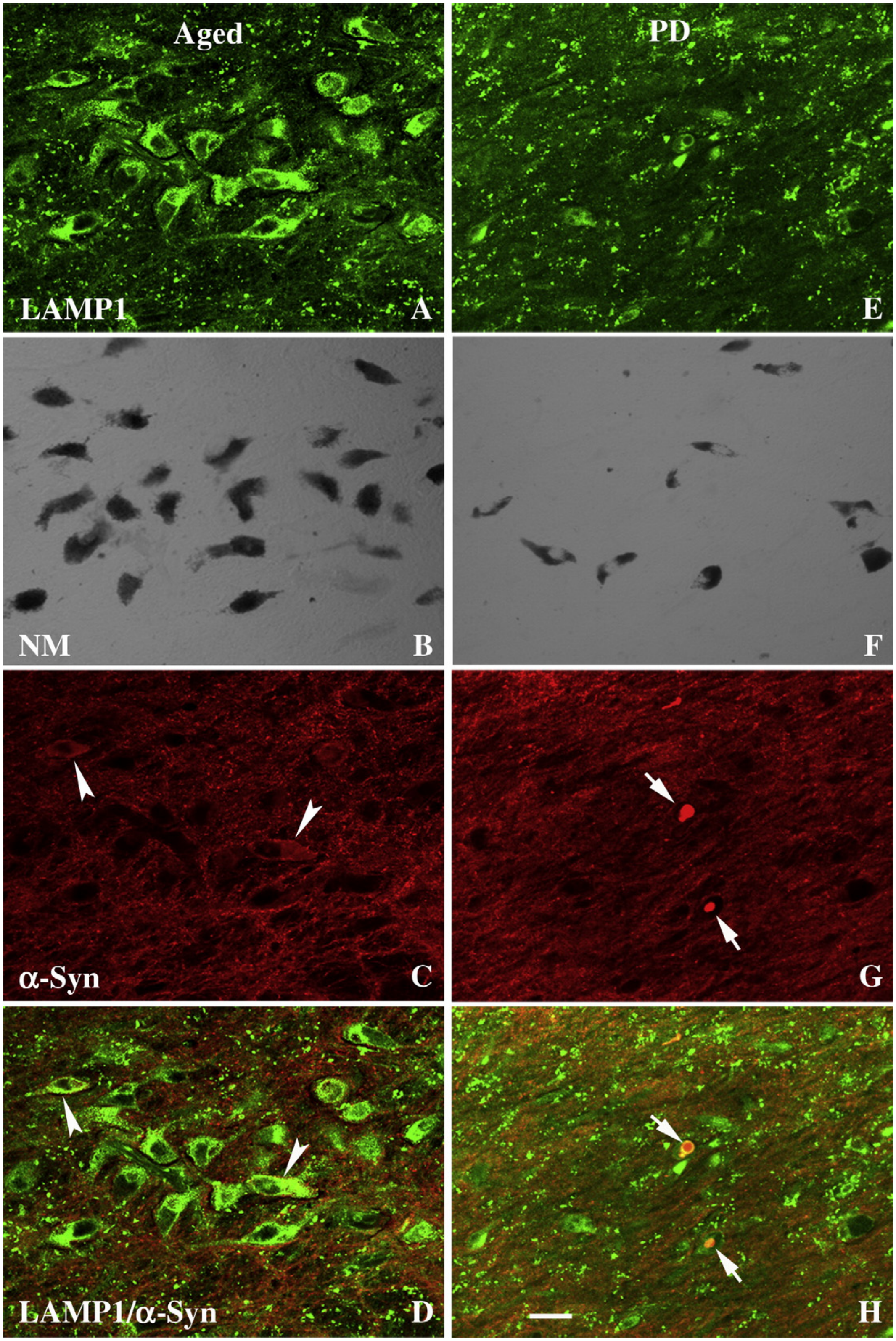

Fig. 1. Laser confocal microscopic images of nigral neuron from age-matched control (A–D) and PD case (E–H), illustrating that LAMP1 (green; A, E), NM (black transparent optics; B, F), α-syn (red; C, G), and co-localization of LAMP1 and α-syn (D, H). Note that LAMP1 immunofluorescence intensity was reduced in nigral neurons without α-syn-ir inclusions (H) and severely diminished in the neurons with α-syn-ir inclusions (arrows; H) in PD. However, the LAMP1 immunofluorescence intensity was unchanged in the neurons with diffuse non-aggregated α-syn-ir cytoplasmic accumulation in age-matched controls (arrowheads; D). Scale bar in H = 60 μm (applies to A–H).

On the uninjected control side, robust immunoreactivity was observed for LAMP1 (Fig. 7D), CatD (Fig. 8D), HSP73 (Fig. 9D), and 20S proteasome (Fig. 10D). LAMP1 and CatD immunostaining profiles were mainly distributed in perikarya. HSP73 immunoreactivity was observed in both the perikarya and processes of nigral neurons. 20S proteasome immunoreactivity was detected in both nucleus and

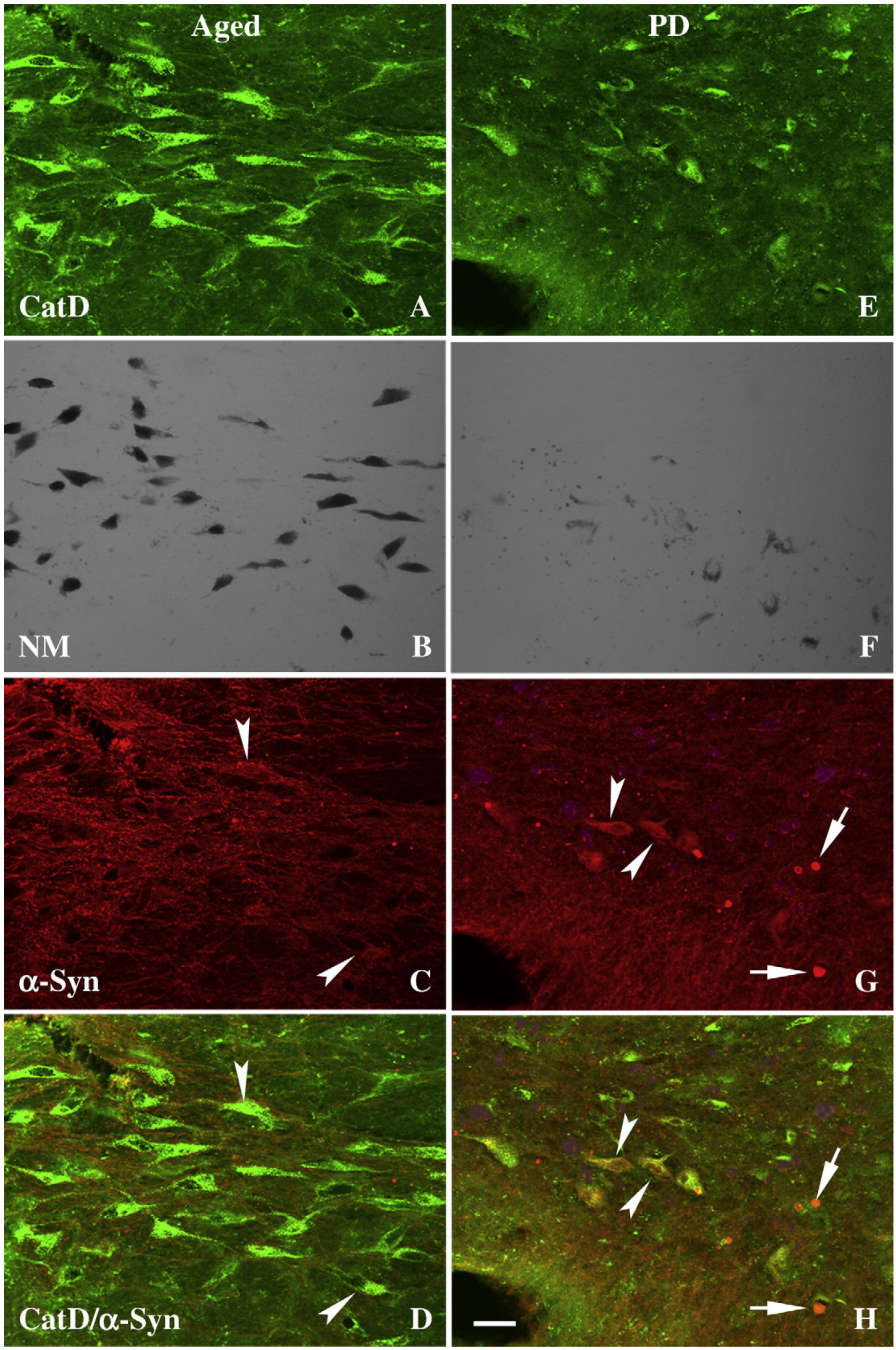

Fig. 2. Confocal images of substantia nigra from age-matched control (A–D) and PD case (E–H), illustrating the CatD (green; A, E), NM (black transparent optics; B, F), α-syn (red; C, G), and co-localization of CatD and α-syn (D, H). Note that CatD immunofluorescence intensity was reduced in the nigral neurons with diffuse α-syn-ir in the cytoplasm (arrowheads; H) and severely diminished in the neurons with α-syn-ir inclusions (arrows; H) in PD. In controls, neurons with diffuse cytoplasmic α-syn-ir (arrowheads; C, D) displayed approximately the same degree of CatD immunostaining intensity as the neurons in which α-syn was not detected in age-matched controls. Scale bar in H = 60 μm (applies to A–H).

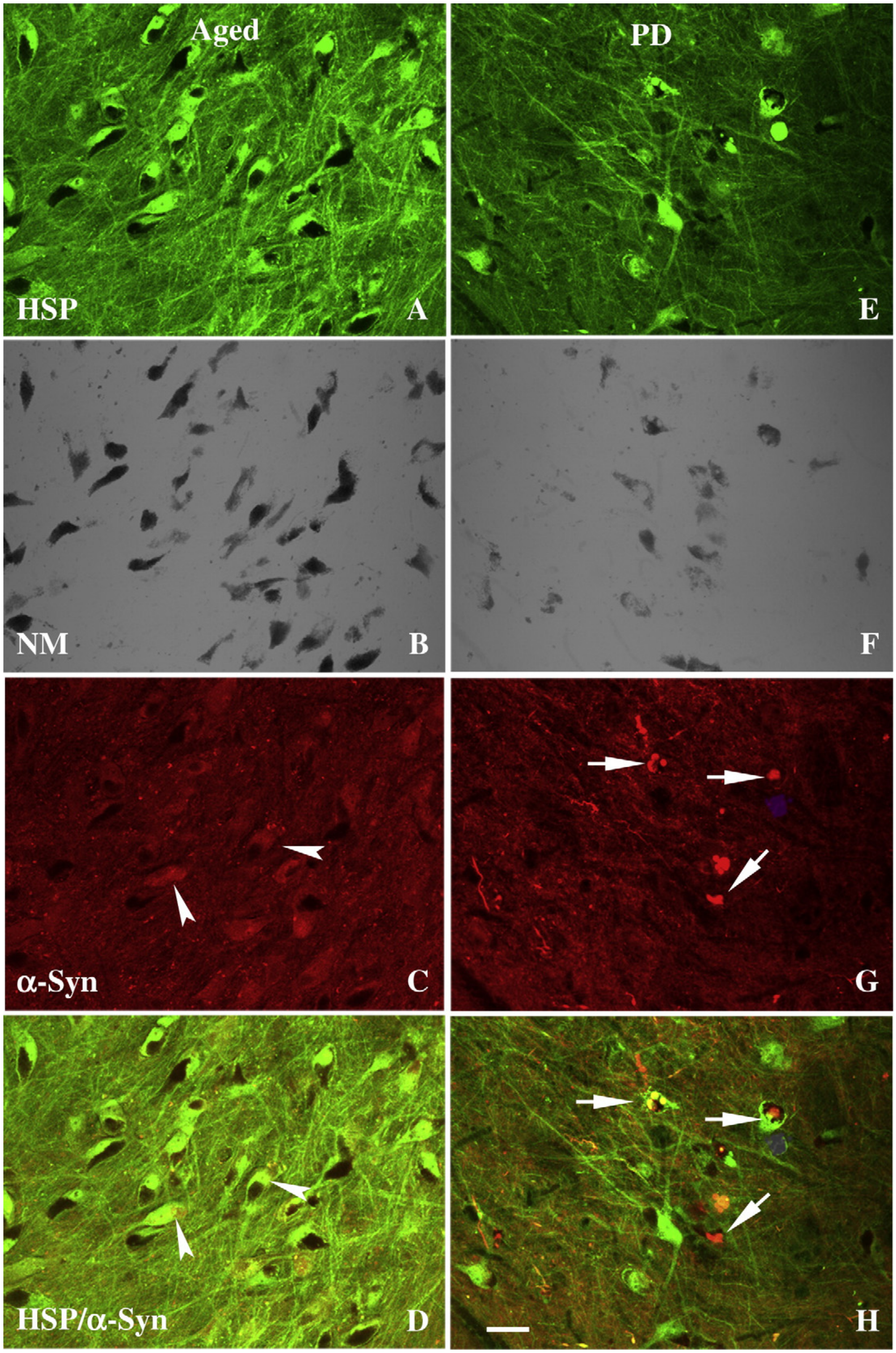

Fig. 3. Confocal microscopic images of substantia nigra from age-matched controls (A–D) and PD case (E–H), illustrating HSP73 (green; A, E), NM (black transparent optics; B, F), α-syn (red; C, G), and co-localization of HSP73 and α-syn (D, H). Note that HSP73 immunofluorescence intensity was severely diminished in the neurons with α-syn-ir inclusions (arrows; H) in PD but not in neurons with diffuse α-syn-ir cytoplasmic accumulation (arrowheads; C, D) in age-matched controls. Scale bar in H = 60 μm (applies to A–H).

perikarya and staining for the 20S proteasome was much stronger in the nucleus than in the perkarya. Robust expression of TH-ir was also observed on the uninjected side (data not shown). All of these finding recapitulate what is seen in the normal control nigra.

On the AAV6-h-A30P injected side, stereological counts revealed 60% loss of TH-ir neurons (data not shown). With regard to LAMP1 (Fig. 7A) and CathD (Fig. 8A) immunoreactivity, some remaining neurons displayed an extensive loss of staining while others

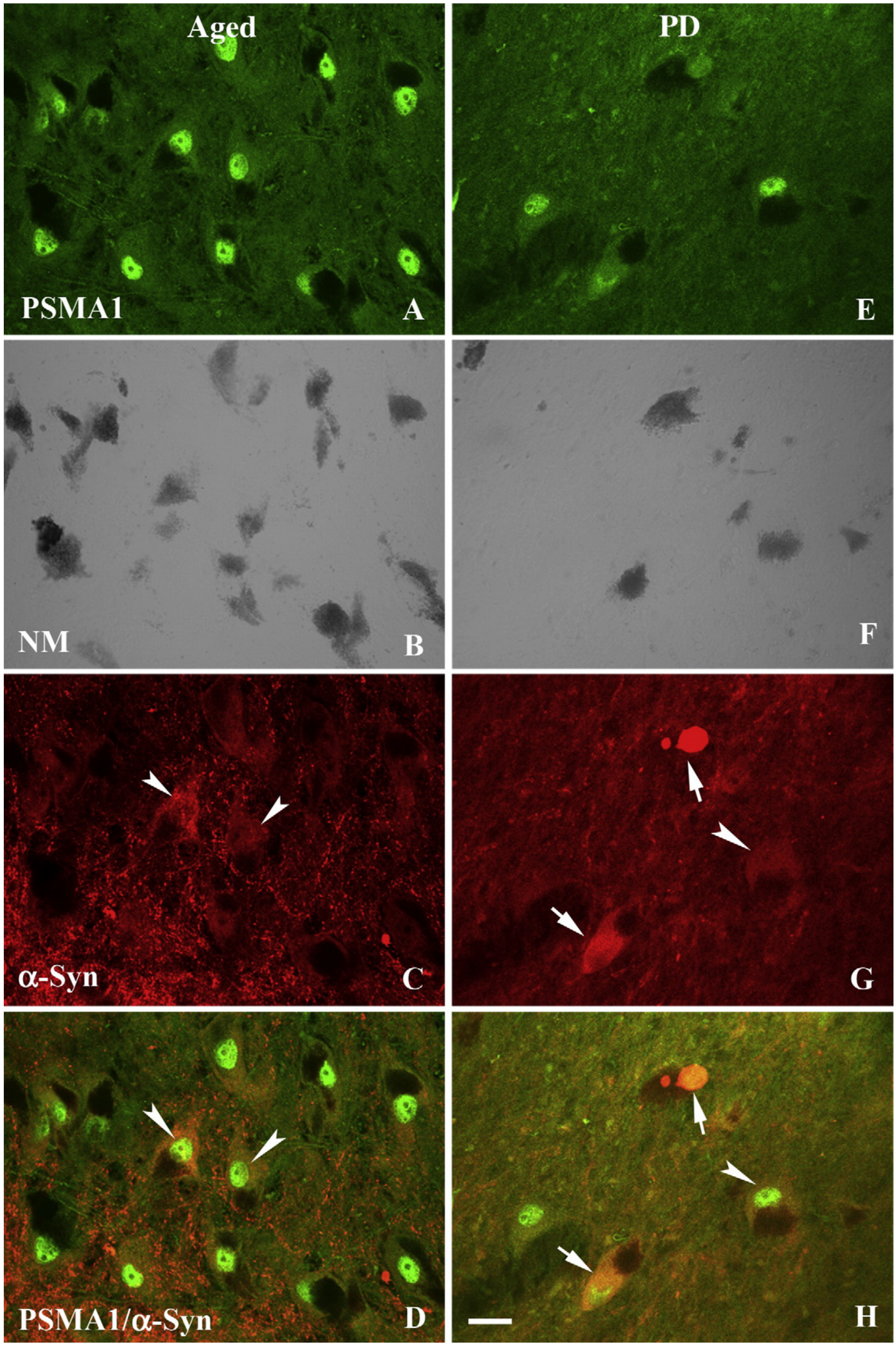

Fig. 4. Confocal microscopic images of substantia nigra from age-matched control (A–D) and PD case (E–H), illustrating immunostaining for 20S proteasome (green; A, E), NM (black transparent optics; B, F), α-syn (red; C, G), and co-localization of 20S proteasome and α-syn (D, H). Note that 20S proteasome immunofluorescence intensity was reduced in the neurons with α-syn-ir inclusions (arrows; H) but not in the neurons with α-syn-ir diffused cytoplasm (arrowheads; C, D). Scale bar in H = 30 μm (applies to A–H).

demonstrated increased expression in the perikarya and processes. For HSP73 (Fig. 9A) and 20S proteasome (Fig. 10A), some of neurons had an extensive loss of immunoreactivity while the others retained their normal immunoreactive profile. Double labeling revealed that there were two groups of neurons in AAV6-h-A30P injected substantia nigra. One class did not display α-syn-ir and these cells intensely

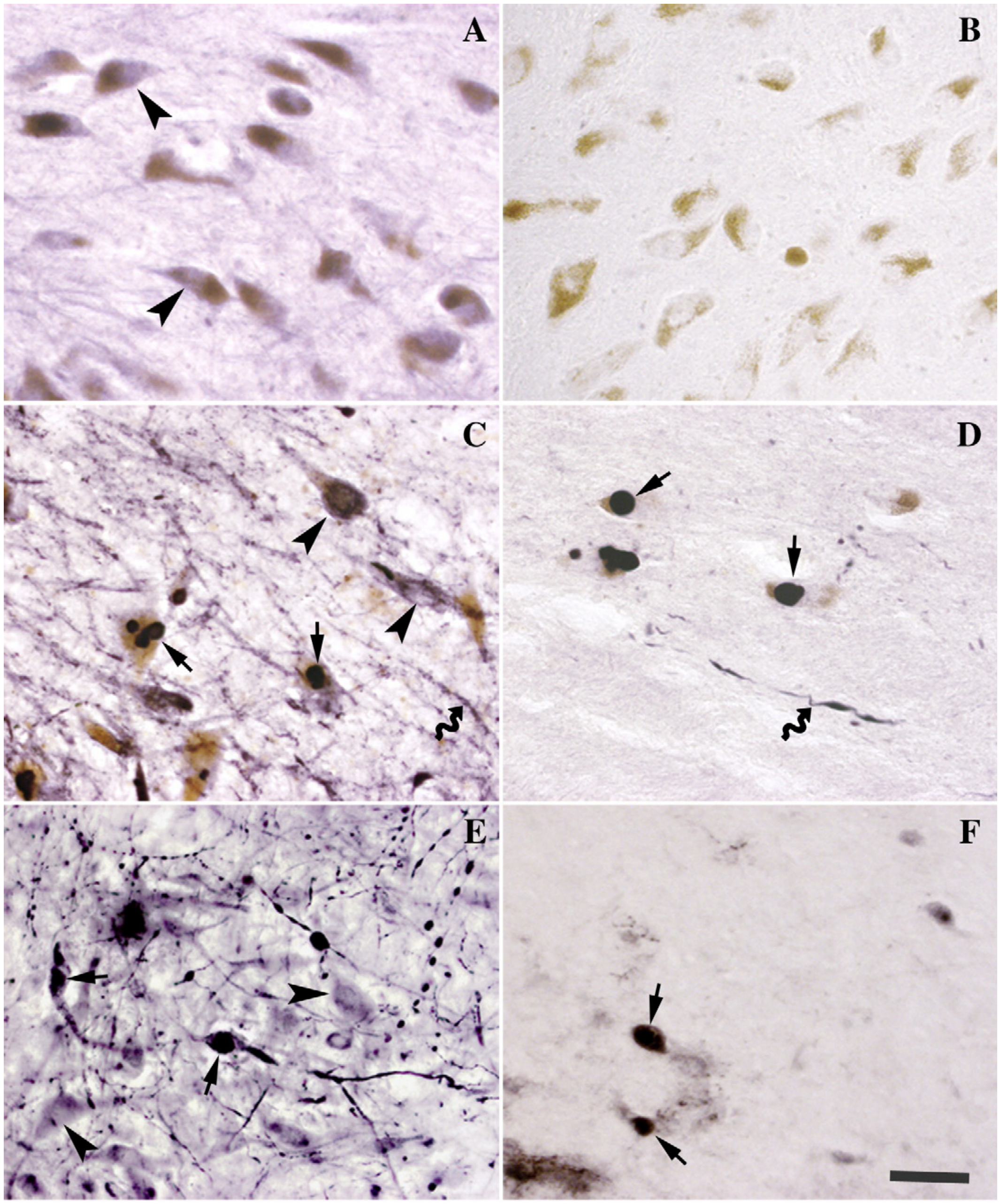

Fig. 5. High-power photomicrographs of nigral sections from normal aged subject (A, B), PD (C, D), and rat with target expression of human mutant α-syn (A30P; E, F), illustrating the patterns of α-syn (A–E) immunoreactivity and resistance to proteinase K treatment (B, D, F). α-syn (A, C, E; arrowheads) immunolabeling observed on the sections without proteinase K treatment was virtually eliminated following proteinase K treated sections (B, D, F). In contrast, immunoreactive inclusions (arrows, C, D) and neurites (curved arrows) were detected on both sections of PD and rat with and without proteinase K treatment, indicating that the inclusions are insoluble. Scale bar = 32 μm in F (applies to A–F).

normally expressed lysosomal and proteasomal markers. The second class co-expressed α-syn aggregates had markedly diminished staining for lysosomal and proteasomal (Figs. 7C–10C). These observations indicate that changes in lysosomal and proteasomal markers are associated with aggregation of α-syn, as was seen in PD.

Lysosomal and proteasomal immunoreactivity in rat substantia nigra with α-syn gene delivery: Quantitative observations

To further test the hypothesis that α-syn aggregates within cells were associated with reduced lysosomal and proteasomal markers,

Table 2
Optic density analysis of LAMP1-ir, CatD-ir, HSP73-ir, and 20S proteasome-ir neurons in substantia nigra.

Group	LAMP1 OD	CatD OD	HSP73 OD	20S proteasome OD
Age-matched control	2069.10 ± 329.52	1809.35 ± 533.47	2604.92 ± 494.56	1660.84 ± 229.87
PD	1261.54 ± 107.77***	1094.64 ± 378.10***	1799.27 ± 376.19***	1172.65 ± 273.28*

Data are means ± SD.
***P<0.001,*P<0.05 compared with age-matched controls; OD: optic density.

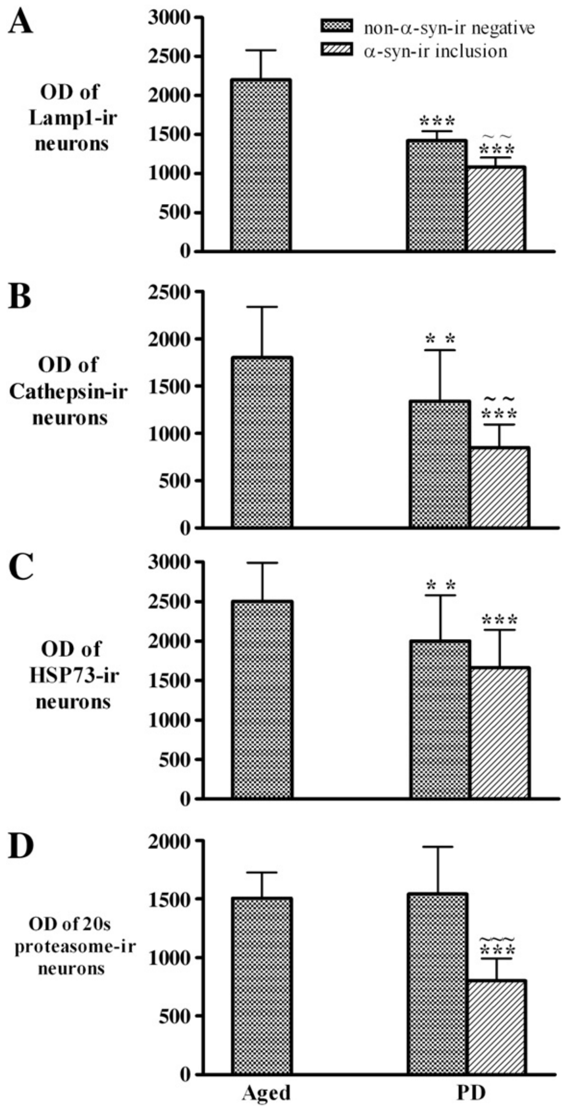

Fig. 6. Histogram showing the optical density (OD) of LAMP1 (A), CatD (B), HSP(C), and 20S proteasome (D) immunofluorescence intensity within substantia nigral neurons in the Parkinson's disease (PD; $n = 10$) and age-matched control (Aged; $n = 10$) groups. The ODs of LAMP1 (A), CatD (B), and HSP(C) immunofluorescence intensity were significantly reduced in both neurons with or without α-syn-ir inclusions in PD relative to age-matched controls. While the neurons with α-syn inclusions displayed more severely decrease of ODs of LAMP1, CatD, and HSP immunofluorescence intensity. The OD of 20S proteasome immunofluorescence intensity was only decreased in the neurons with α-syn-ir inclusions. (***P*<0.05; ****P*<0.01 compared with age-matched controls and ~~*P*<0.05; ~~~*P*<0.01 related to neurons with α-syn-ir inclusion in PD; Data are means ±SD.).

we quantified fluorescence intensity measurements for lysosomal and proteasomal markers independently in nigral neurons with and without α-syn accumulation as described above. The OD measurements of LAMP1 (Fig. 11A), CatD (Fig. 11B), HSP73 (Fig. 11C), and 20S proteasome (Fig. 11D) were significantly decreased in the neurons with α-syn-ir aggregates (*P*<0.05–0.01) but not in nigral neurons without these deposits (*P*>0.05). Interestingly LAMP1 and CatD OD were significantly increased in nigral neurons displaying diffuse, but non-aggregated, α-syn-ir relative to the control side (Fig. 11B). HSP73 and 20S proteasome levels in nigral neurons with diffuse α-syn immunoreactive deposits were similar to intact neurons.

Discussion

In the present study we examined the relationship between α-syn aggregates and the expression of markers of UPS and lysosomal function in patients with PD. We found that immunoreactivity for the lysosomal markers LAMP1 and CatD were significantly decreased in remaining nigral DA neurons in comparison to age-matched controls, and these reductions were significantly more pronounced in neurons that contained α-syn positive inclusions. Similar findings were observed in PD cases with regard to the expression of HSP73 which was significantly decreased in remaining nigral neurons, and most severely reduced in neurons with α-syn inclusions. This is relevant since heat shock proteins serve to facilitate the refolding of misfolded proteins and act as chaperones for lysosomal and proteosomal systems. Immunoreactivity for the 20S proteasome was also markedly reduced in dopamine neurons in the PD nigra, but only in cells containing α-syn-ir inclusions, with no reduction in nigral cells that have diffuse or no α-syn expression. These qualitative observations were confirmed by quantitative OD measurements. This pattern was recapitulated in rats with targeted over-expression of mutant α-syn in the substantia nigra which resulted in dopamine cell loss with inclusions. Our findings indicate that aggregates of α-syn are associated with alterations in proteasome and lysosomal markers that are not found, or are markedly less pronounced, in dopamine neurons that do not contain α-syn inclusions. These findings are in keeping with the notion that Parkinson's disease is related to proteolytic stress and that the accumulation and aggregation of α-syn is a critical factor in this process. However it should be noted that while the most parsimonious explanation of the present finding is reduced protein expression, it remains possible that the studied proteins might be reduced due to reduced detectability due to aggregation.

Accumulation and aggregation in the form of Lewy bodies and Lewy neurites is a pathological hallmark of PD (Spillantini et al., 1997). α-syn is a naturally unfolded protein that is prone to misfold and aggregate when present in high concentrations or in a mutant form. The protein is normally cleared by way of both chaperone-mediated autophagy and the UPS (Lee et al., 2004; Pan et al., 2008; Vogiatzi et al., 2008; Paxinou et al., 2001; Cuervo et al., 2004; Xilouri et al., 2008; Liu et al., 2003; 2005; Meijer and Codogno, 2004). Accumulation of α-syn could result from either increased production or impaired clearance of the protein, or both. There is evidence to suggest that this might occur in PD.

Familial forms of PD are associated with mutations in α-syn which promote misfolding and aggregation of the protein (Polymeropoulos et al., 1997; Krüger et al., 1998). Importantly, familial PD cases have also been reported with duplication or triplication of wildtype α-syn (Singleton et al., 2003; Nishioka et al., 2006), indicating that excess levels of even the normal protein can lead to accumulation and aggregation, perhaps because they overwhelm the capacity of the normal UPS and autophagy systems to clear the protein. In the laboratory, over-expression of mutant or wildtype α-syn in transgenic mice (Masliah et al., 2000) and *Drosophila* (Feany and Bender, 2000) has been shown to cause dopamine neuronal degeneration with inclusion bodies. Similarly, targeted over-expression of α-syn in the nigrostriatal system of rats and nonhuman primates induces dopamine neuronal degeneration with inclusions and mimics several of the behavioral features of PD (Kirik et al., 2002, 2003). Collectively, these findings indicate that excess production of α-syn can induce a PD syndrome.

There is also evidence indicating that PD is associated with impaired clearance of α-syn. While the lysosomal system has not been well studied to date, there UPS systems appears functional in the SNc in patients with sporadic PD. In these patients, there is reduced expression of alpha, but not beta, subunits of the 20S proteasome with a reduction in the activity of each of the proteasomal enzymes (McNaught et al., 2003). Further, compensatory up-regulation in the PA700 proteasome activator does not occur within the SNc but does occur in brain regions that do not degenerate in PD (McNaught et al., 2003). In the laboratory, in

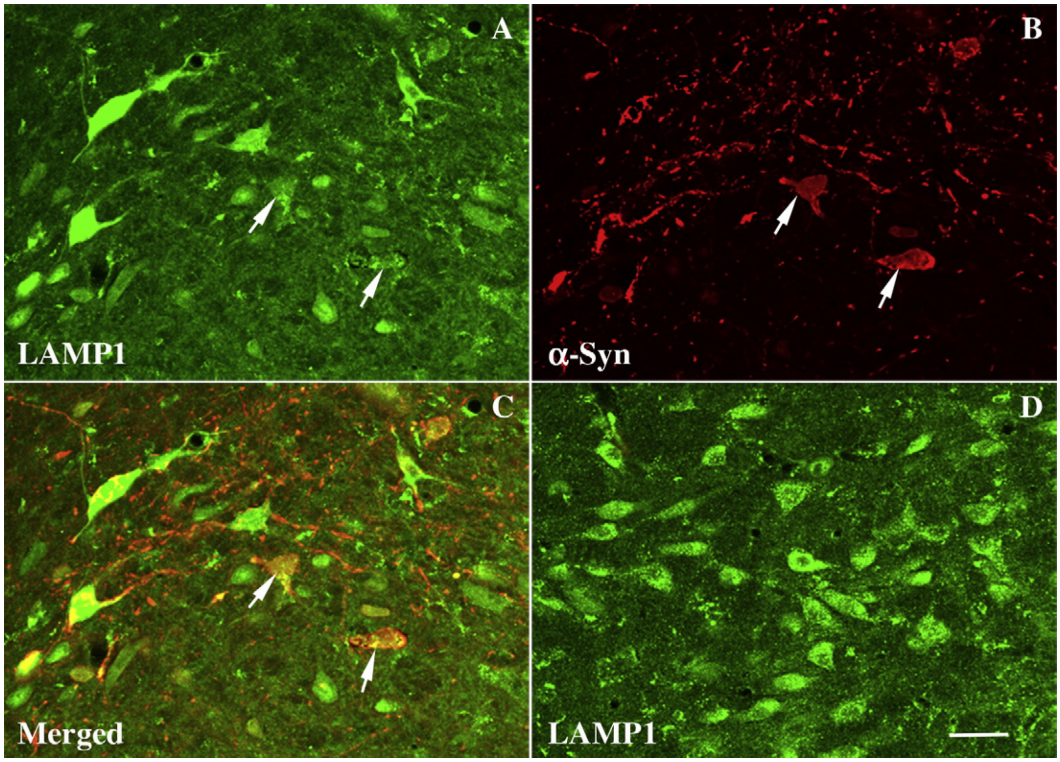

Fig. 7. Laser confocal microscopic images of substantia nigra from rat with targeted expression of human mutant (A30P) α-syn (A–C) and intact side (D), illustrating the intensity of LAMP1 (green; A, D) and α-syn (red; B) immunofluorescence and co-localization of LAMP1 and α-syn (C). Note that the LAMP1 immunofluorescence intensity (arrows) was severely diminished in neurons that express α-syn inclusions. Scale bar in D = 30 μm (applies to A–D).

vitro studies as well as direct infusions of proteasome inhibitors into the ventral midbrain lead to dopamine neuronal degeneration coupled with the formation of inclusion bodies that stain positively for α-syn (McNaught et al., 2002a,b). Collectively, these findings indicate that impaired clearance of α-syn could also lead to inclusion body formation and a PD syndrome.

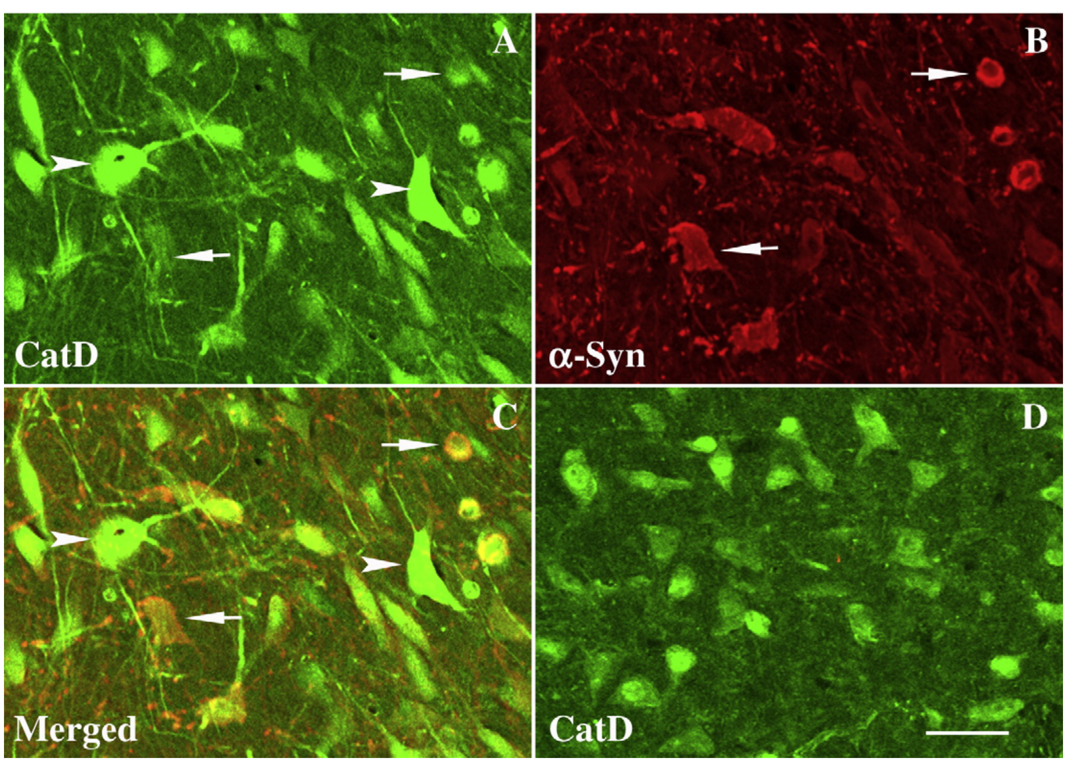

Fig. 8. Laser confocal microscopic images of substantia nigra from rat with target expression of human mutant (A30P) α-syn (A–C) and intact side (D), illustrating the intensity of CatD (green; A, D) and α-syn (red; B) immunofluorescence and co-localization of CatD and α-syn (C). Note that the CatD immunofluorescence intensity was severely diminished by α-syn inclusions (arrows) but increased in the neurons without α-syn inclusions (arrowheads). Scale bar in D = 30 μm (applies to A–D).

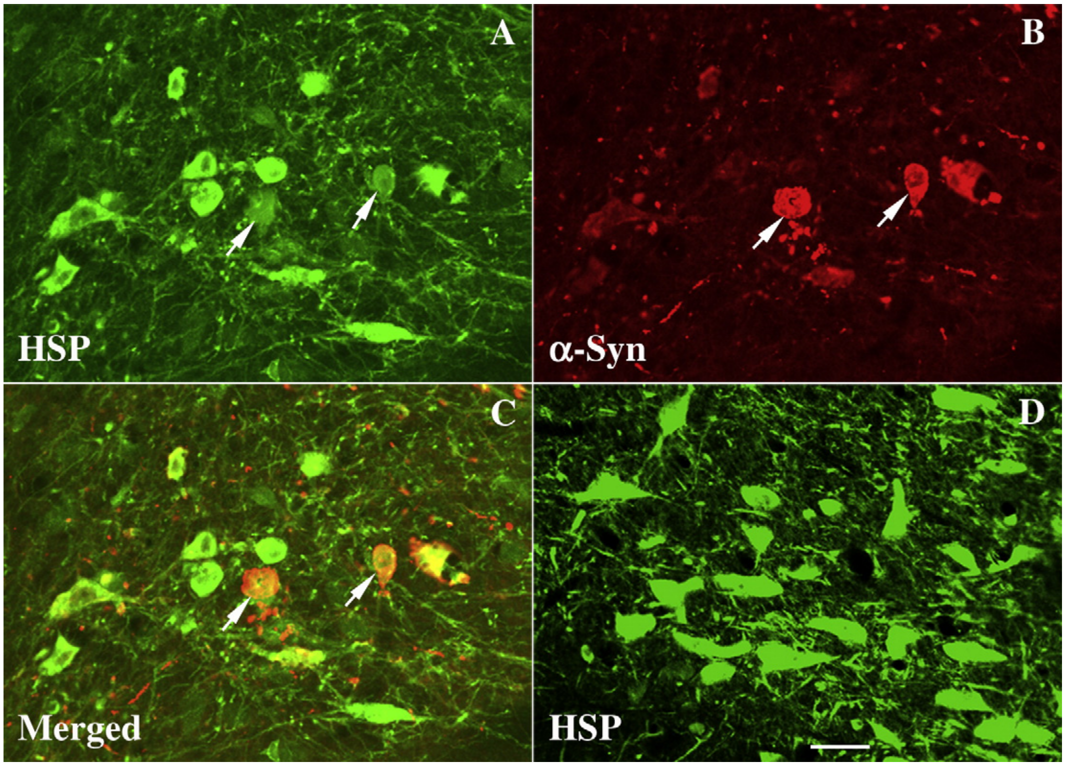

Fig. 9. Laser confocal microscopic images of substantia nigra from rat with target expression of human mutant (A30P) α-syn (A–C) and intact side (D), illustrating the intensity of HSP73 (green; A, D) and α-syn (red; B) immunofluorescence and co-localization of HSP73 and α-syn (C). Note that the HSP73 immunofluorescence intensity was severely diminished by α-syn inclusions (arrows). Scale bar in D = 30 μm (applies to A–D).

Aggregated α-syn can contribute to the development of proteolytic stress regardless of the mechanism responsible for its formation. In experimental systems, α-syn aggregation has been shown to inhibit lysosomal and proteasomal functions and to thereby impair the capacity of these systems to clear α-syn as well as other unwanted proteins (Bennett et al., 2005; Cuervo et al., 2004 Tanaka et al., 2001b;

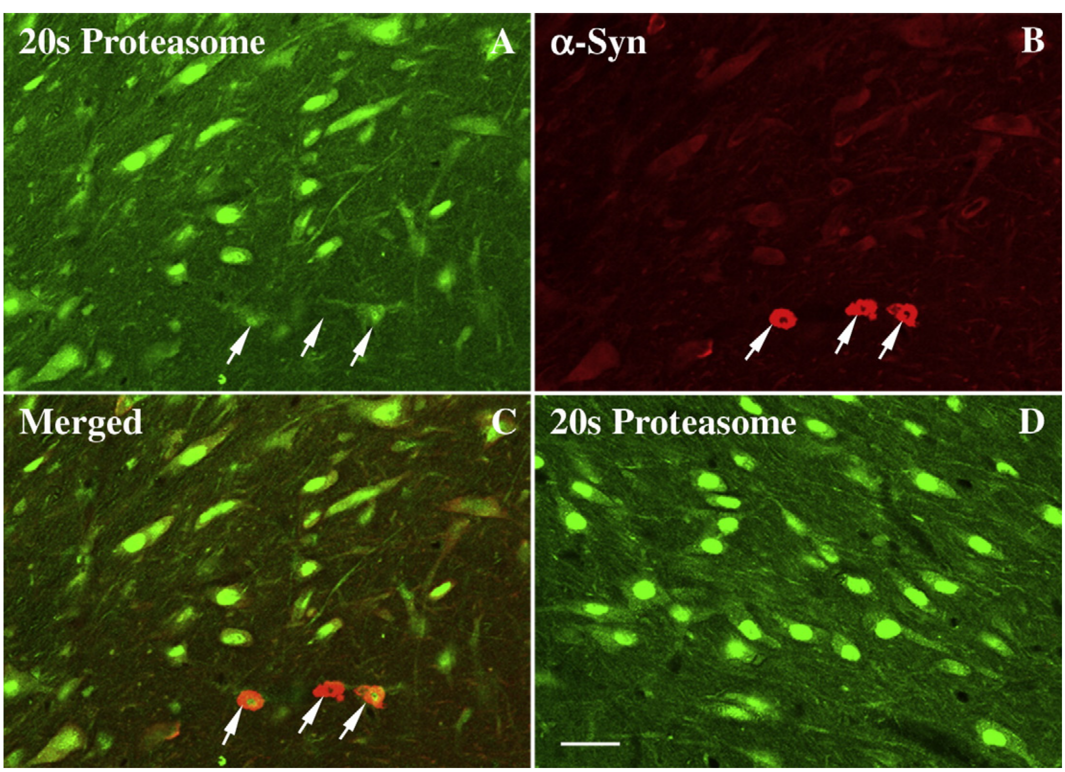

Fig. 10. Laser confocal microscopic images of substantia nigra from rat with target expression of human mutant (A30P) α-syn (A–C) and intact side (D), illustrating the intensity of 20S proteasome (green; A, D) and α-syn (red; B) immunofluorescence and co-localization of 20S proteasome and α-syn (C). Note that the 20S proteasome immunofluorescence intensity is severely diminished in the presence of α-syn inclusions (arrows). Scale bar in D = 30 μm (applies to A–D).

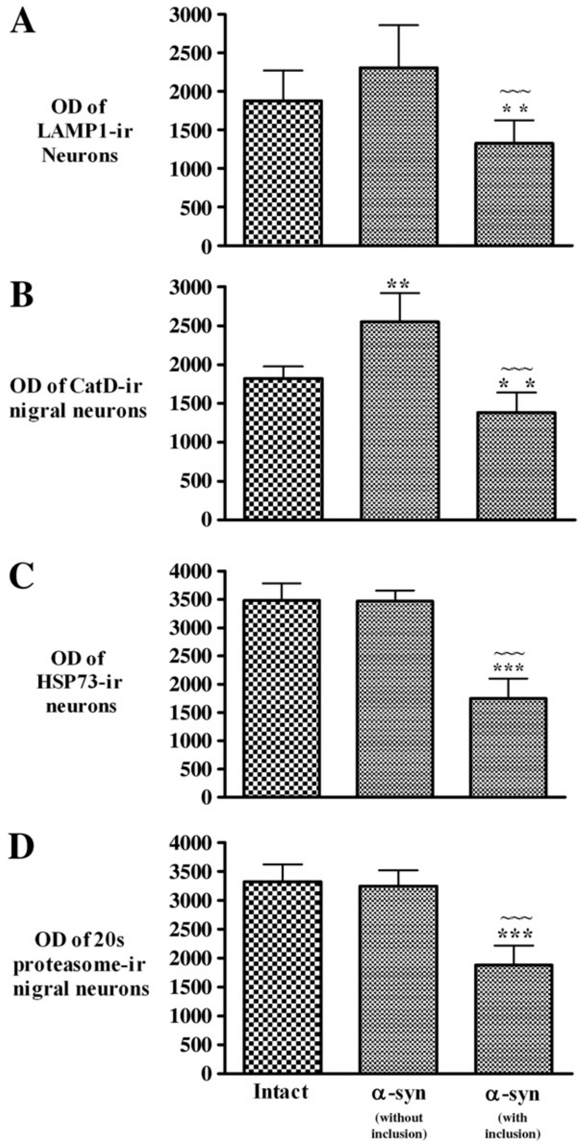

Fig. 11. Histogram showing that the optical density (OD) of LAMP1 (A), CatD (B), HSP73 (C), and 20S proteasome (D). Immunofluorescence intensity was significantly decreased in neurons with α-syn-ir inclusions compared to neurons without α-syn-ir inclusions on the same side and to control neurons on the intact side in rats with targeted expression of human mutant (A30P) α-syn. However, the OD of CatD (B) was significantly increased in the neurons without α-syn-ir inclusions. (***P*<0.05; ****P*<0.01 compared with the intact control and ~~~*P*<0.01 compared with the neurons without α-syn-ir inclusions; Data was means ± SD.).

Snyder et al., 2003; Martinez-Vicente et al., 2008). Thus, a vicious cycle could develop in which α-syn accumulation due to either increased production or impaired clearance could lead to aggregation of the protein, inhibition of the lysosome/proteasome system, further accumulation/aggregation of α-syn and other unwanted proteins, the development of proteolytic stress. Our findings of the coexistence of α-syn aggregates with reduced expression of markers of the lysosomal and UPS systems raise the possibility that this vicious cycle exists in PD (Figs. 2H, 4C, 8C, 10C).

The significance of α-syn positive inclusions or Lewy bodies in PD patients remains a subject of debate. We have argued that they represent a form of aggregation that develops in response to excess levels of misfolded or unwanted proteins in order to facilitate their clearance (Olanow et al., 2004). This would imply that the inclusions in PD are protective, and represent a compensatory response to a state of proteolytic stress resulting from the inability of the lysosomal and proteasomal systems to clear excess levels of unwanted protein. Indeed, it has been shown experimentally that the formation of inclusion bodies only occurs after impairment of the proteasomal system by nuclear or cytoplasmic protein aggregates (Bennett et al., 2005). Experimentally, inhibition of microtubular transport with impairment in the capacity to form inclusions is associated with more severe neuronal degeneration (Olanow et al., 2004; Kawaguchi et al., 2003), and in PD Lewy bodies have not been shown to be associated with alterations in cell morphology or markers of cell stress, in comparison to dopamine neurons without inclusions (Tompkins and Hill, 1997).

To determine if excessive α-syn alone is sufficient to generate this cycle, we used an AAV6 viral vector to induce targeted over-expression of α-syn in dopamine neurons in the SNc of rodents. Six weeks following the injections, both diffuse and aggregated α-syn positive inclusions were observed within nigral neurons. Diminished expression of proteasomal and lysosomal markers were found in neurons with α-syn inclusions compared to those with diffuse or no α-syn accumulation, a pattern identical to what we observed in the PD brain. Thus the vicious cycle of degenerative events detailed above can be initiated solely by α-syn over-expression. This is likely to reflect what occurs in PD cases with α-syn duplications or triplications and possibly those cases with mutations in the α-syn protein. These patients with α-syn mutations, however, constitute a small minority of PD cases. We hypothesize that this same cycle could be induced by proteasomal/lysosomal dysfunction caused by genetic or environmental factors in sporadic PD.

Dopamine neurons in aged human control brains in our study also contain diffuse intracellular α-syn in some SNc neurons, but not aggregated α-syn positive inclusions as seen in PD. We have previously demonstrated that there is an age-related accumulation of α-syn (Chu and Kordower, 2007). We now find that the accumulation of α-syn in elderly controls is not aggregated and is not associated with alterations in markers of the lysosome or proteasome systems in comparison to cells which do not contain α-syn at all. These observations suggest that in normal aging compensatory mechanisms are present which are sufficient to prevent the aggregation of α-syn and consequent damage to lysosomal and proteasomal systems. Indeed, a similar compensatory response might be seen in AAV6-h-A30P injected rats. In these animals, lysosomal markers were increased in nigral neurons with diffuse α-syn expression. We speculate, however that the accumulation of α-syn by either increased production or impaired clearance ultimately cannot be adequately compensated in PD, resulting in a vicious cycle involving continued accumulation of the protein, formation of protein aggregates, damage to the lysosomal and proteasomal systems, further protein accumulation, and ultimately degeneration of dopamine neurons.

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